# DELHI TECHNICAL CAMPUS

Affiliated to GGSIPU, New Delhi NAAC-A++ Greater Noida

Session 2025 - 26

## ROUTE SHARE : An Intelligent Carpool System



Kumar Tejaswa


Shobhit Shukla


## Introduction

Managing daily travel in cities can be stressful, costly, and inefficient, especially when people commute on similar routes but travel separately. The Route Share Carpool System solves this issue by offering a smart platform that connects drivers and passengers moving in the same direction. Whether traveling to college, office, or long-distance locations, it streamlines the ride-sharing process with automation and intelligent decision-making.
The system uses modern web technologies and AI-powered route optimization to match users accurately, suggest the most efficient routes, calculate travel costs, and enable secure digital payments through Stripe. Real-time tracking via Socket.IO provides live location updates, while JWT authentication ensures safety and reliability for every user. Ride creation, searching, booking, and monitoring are handled seamlessly without manual coordination.
Overall, the system aims to make travel more affordable, reduce congestion, promote sustainability, and save time for users who commute regularly on fixed or flexible routes, while delivering a smooth and intelligent ride-sharing experience.

## Objectives

The primary objective of the Route Share Carpool System is to offer users a reliable and efficient way to share rides using intelligent route optimization and modern web technologies.
This system aims to:

- Reduce individual travel costs by allowing users to share rides along similar routes.
- Enhance efficiency by using AI-based route planning to minimize time, distance, and fuel consumption.
- Improve safety and reliability through secure authentication, verified profiles, and ride history tracking.
- Increase accessibility of rides through real-time location tracking and live route updates.
- Offer a user-friendly web interface that requires no technical expertise to operate.

By achieving these goals, the system empowers users to commute more affordably and efficiently while supporting sustainable and eco-friendly transportation practices

## Feasibility Study and Tool Used

The Route Share Carpool System is developed using reliable, scalable, and efficient technologies that ensure smooth real-time tracking, accurate route calculations, secure authentication, and seamless user interaction. The tools and technologies used include:

- Frontend: React, Vite, and Tailwind CSS for a fast, responsive, and visually clean interface suitable for both mobile and desktop screens.
- Backend: Node.js and Express.js to manage authentication, ride creation, booking logic, APIs, and system communication.
- Database: MongoDB Atlas for securely storing user profiles, ride details, bookings, and trip histories in a flexible cloud environment.
- AI Integration (NX8 Model): Hugging Face Mistral 8x7B (NX8) model to generate intelligent ride explanations, reasoning, and context-aware suggestions for better ride matching.
- Routing: OpenRouteService (ORS) API to compute optimized routes, distances, travel durations, and ride compatibility.
- Payments: Stripe API for secure and encrypted online transactions, generating instant confirmations for ride bookings.

## System Architecture

1. Frontend
- React + Tailwind for a responsive UI.
- Handles login, ride creation, and tracking.

2. Backend
- Node.js & Express manage auth, rides, and APIs.
- Connects to MongoDB for all data.

3. AI & Routing
- ORS API for optimized routes.
- NX8 (Mistral 8x7B) for AI explanations.

4. Real-Time Tracking
- Socket.io sends live GPS updates.

5. Database
- MongoDB Atlas stores users, rides, and bookings.

## Design and Implementation

1. Authentication Module
- Implements secure login and registration using JWT so only verified users access the system.
- Stores encrypted passwords and user details in MongoDB to protect sensitive information.
- Validates user tokens on every request to ensure a secure ride creation and booking environment.

2. Ride Management
- Drivers create rides by entering route, timing, seat availability, and pricing.
- Passengers search for rides using source and destination filters.
- Displays all available rides in real time with driver, fare, seats, and route information.

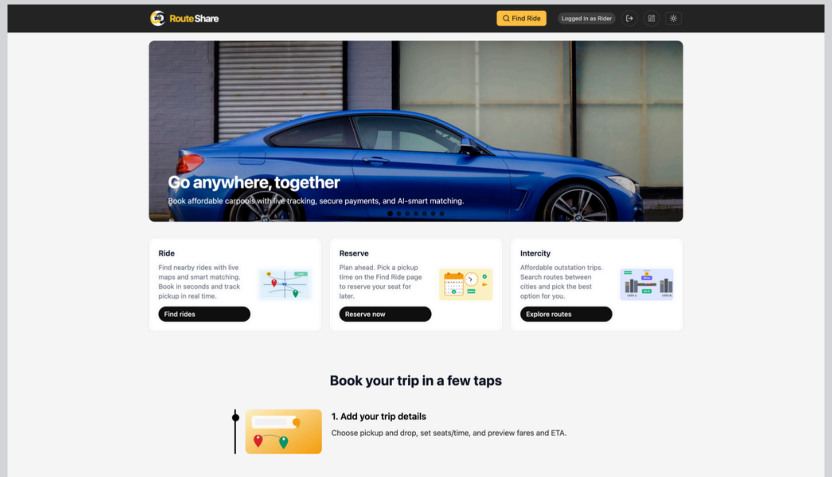
3. Route Optimization
- Uses OpenRouteService (ORS) API to generate shortest and most efficient routes.
- Calculates distance, travel duration, and estimated arrival time with high accuracy.
- Compares user routes and suggests rides that best match passenger travel paths.

4. Real-Time Tracking
- Socket.io provides continuous GPS updates throughout the ride.
- The map updates automatically so passengers can monitor the driver's live location.
- Improves coordination by showing real-time arrival and route progress.

5. AI Explanations
- Uses the Mistral 8x7B (NX8) model to provide intelligent reasoning for ride matching.
- Generates clear explanations to help users understand why a ride is recommended.

6. Payment Integration
- Stripe API processes secure digital payments for confirmed rides.
- Encrypts card details and generates instant payment confirmations.
- Stores transaction IDs and booking status for easy ride history review.

## Result




## Conclusion

The Route Share Carpool System successfully addresses the challenges of daily commuting by offering a smart, organized, and real-time platform for sharing rides. By combining AI-powered route optimization, secure authentication, live GPS tracking, and digital payments, the system provides a seamless experience for both drivers and passengers. It improves coordination, enhances safety, and ensures accurate route planning with reliable real-time updates. Overall, the system delivers a modern, user-friendly, and sustainable solution that simplifies ride-sharing and makes daily travel more efficient for regular commuters.

## Batch Details

| Student's Name | Enrollment No. |
|---|---|
| Kumar Tejaswa | 14718002722 |
| Shobhit Shukla | 04618002722 |

Supervisor Name: Mr Vibhav Ranjan